Unveiling the Complexities of Artificial Intelligence

Addressing Limitations, Challenges, and Ethical Concerns

ABSTRACT – Artificial Intelligence (AI) technologies lack morality, as well as other fundamental characteristics inherently related to the meaning of the word "intelligence". AI systems miss common sense, they show difficulties in handling contextual elements or semantic meaning, and emotions are completely absent in their understanding. Due to those deficiencies, entrusting AI with decisions related to human lives becomes a matter of concern. This essay aims to demonstrate that AI systems should not be in charge of ethical matters, morality and human affairs. In our discussion, we prove that the technical structure of AI algorithms prevents them from holding responsibility for decisions concerning human lives.

KEYWORDS – artificial intelligence, morality, ethics, bias, discrimination.

Introduction

While being the revolutionary technology that enable machines to think, learn and perform tasks simulating human intelligence, Artificial Intelligence systems cannot simulate the complexity of human life. After having thoroughly explained the different types of AI, as well as the data gathering and understanding processes core of their technology, this essay will focus on the inherent bias and limitations of AI systems, to discover the many ethical problems algorithms may lead to. There will also be explored other external factors contributing to enhance the ambiguity around the understanding of AI functioning mechanisms, such as Big Tech narratives and anthropomorphization.

1. Understanding Artificial Intelligence

There are several definitions for Artificial Intelligence. It has been described as "the study of the computations that make it possible to perceive, reason, and act"[1]; it can be seen as "the automation of intelligent behaviour"[2] driven by a general "study of intelligent agents"[3] both biological and artificial. More generally, two are the angles from which AI can be identified. The former takes in account the ability to perform tasks the same way intelligent beings, more specifically humans, do. The latter considers AI as a system capable of perceiving a certain context and act to reach a certain goal (which could be imposed, self-generated or learned).[4] The most complete definition, however, is that of AI as a non-human intelligence algorithmically[5] set to replicate human

[1] De Spiegeleire, Stephan, Matthijs Maas, and Tim Sweijs. "What Is Artificial Intelligence?", *Artificial Intelligence and The Future of Defense: Strategic Implications for Small-and Medium-Sized Force Providers*. Hague Centre for Strategic Studies, 2017, pp.27-28. http://www.jstor.org/stable/resrep12564.7.
[2] Ibidem.
[3] Ibidem.
[4] Manyika, James. "Getting AI Right: Introductory Notes on AI & Society." *Daedalus*, vol. 151, no. 2, 2022, p.8. https://www.jstor.org/stable/48662023.
[5] "An algorithm is a self-contained step-by-step set of operations that computers and other 'smart' devices carry out to perform calculation, data processing, and automated reasoning tasks. Increasingly, algorithms implement institutional decision-making based on analytics, which involves the discovery, interpretation, and communication of meaningful patterns in data. Especially valuable in areas rich with recorded information, analytics relies on the simultaneous application of statistics, computer programming, and operations research to quantify performance." See: Kossow, Niklas,

mental skills, and capable of recognize pattern, understanding natural languages and responsively learning from any given background (data), and, ultimately, formulating logic response.[6]

Modern AI employs advanced computational methods to automatize and simplify specific tasks, resulting in matching or surpassing human cognitive ability. A broad distinction can be made, to classify and overlook at three ranks of AI. An enumeration will follow.[7]

1. Artificial Narrow Intelligence (ANI), or "Narrow AI", is the application-specific kind of AI, and refers to machine intelligence that can complete specific assignments at (or above) human intelligence levels. Examples include game systems, such as IBM's Deep Blue[8] (for playing chess), Watson (for the game of "Jeopardy!") and Google's AlphaGo (for "Go"); this is also the case of high-frequency trading algorithms, as well as specialized automatic systems like Google Translate, spam filters, chat-bots, and smart assistants (like Siri, Alexa and Cortana) among others.
2. On the other hand, Artificial General Intelligence (AGI) or "strong AI", represents the perfect match with human performance across a wide range of tasks. It encompasses the ability to excel in any given chore, with the same level of proficiency as humans.
3. Lastly, Artificial Superintelligence (ASI) denotes AI that surpasses human intelligence across all performances. It signifies a level of Machine Learning (ML) that exceeds human capabilities in any given domain. [9]

The progress of AI significantly relies on the advancement of computer science research and big data development. Alongside its hardware technology, the true beating heart of AI is the algorithmic power whereby it can analyse large datasets. In the process of learning and understanding data, AI is capable of making statistical inferences and predictions comparable to human.

Svea Windwehr, and Matthew Jenkins. "Algorithmic Transparency and Accountability." Transparency International, 2021, p.4. http://www.jstor.org/stable/resrep30838.

[6] De Spiegeleire, Stephan, Matthijs Maas, and Tim Sweijs. "What Is Artificial Intelligence?", *Artificial Intelligence and The Future of Defense: Strategic Implications for Small-and Medium-Sized Force Providers*. Hague Centre for Strategic Studies, 2017, pp.27-28. http://www.jstor.org/stable/resrep12564.7.

[7] Ivi, p.30.

[8] "Deep Blue had an impact on computing in many different industries. It was programmed to solve the complex, strategic game of chess, so it enabled researchers to explore and understand the limits of massively parallel processing. This research gave developers insight into ways they could design a computer to tackle complex problems in other fields, using deep knowledge to analyse a higher number of possible solutions. The architecture used in Deep Blue was applied to financial modelling, including marketplace trends and risk analysis; data mining—uncovering hidden relationships and patterns in large databases; and molecular dynamics, a valuable tool for helping to discover and develop new drugs." See: IBM100 - Deep Blue.

[9] AGI, as well as ASI, remains non-existent. This is mostly due to the fact that they necessitate more advanced machine learning systems to attain a level of capability at least comparable to humans. The debate regarding the feasibility of achieving artificial general intelligence persists. Should AGI become reality, the implications for humanity would be great, both encompassing beneficial and concerning aspects. See: Manyika, James. "Getting AI Right: Introductory Notes on AI & Society." *Daedalus* 151, no. 2, 2022, p.12. https://www.jstor.org/stable/48662023.

Based on the progress of AI's potential, DARPA[10] outlined three waves of AI's evolution in its capabilities. Starting from the Notional Intelligence Scale, four levels of intelligent capacity are assessed.[11]

1. Perceiving: the detection and determination of global circumstances. AI is able to perceive rich, complex and subtle information.
2. Learning: acquiring knowledge and adapting to diverse scenarios, within an environment.
3. Abstracting: deducing meaning from discovered matters and applying it at another level, to create new meanings.
4. Reasoning: explaining logically and making/planning consistent decisions.

We have to take the four factors above into account, to explain the distinction among the three waves of AI, separated in "Handcrafted Knowledge", "Statistical Learning" and "Contextual Adaptation".

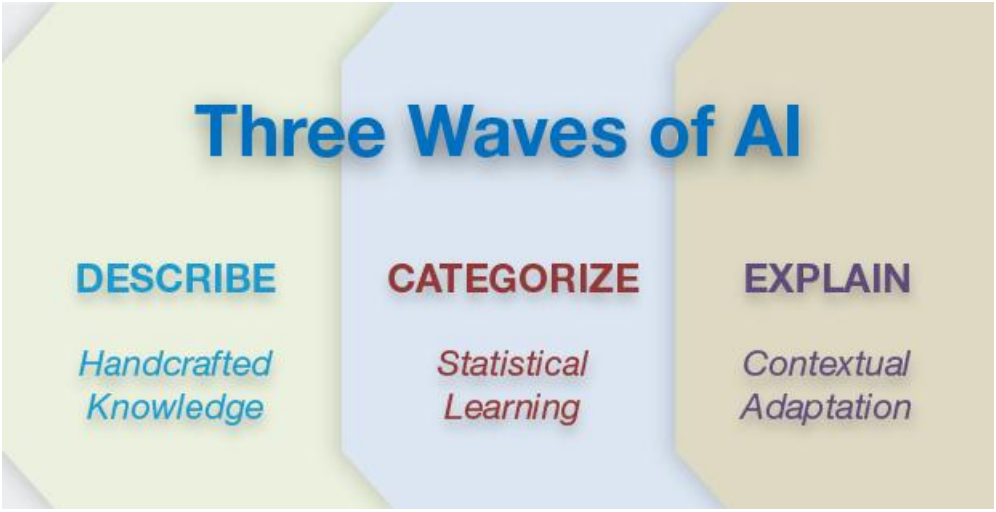


Figure 1, Three Waves of AI.

[10] The Defence Advanced Research Projects Agency (DARPA) of the US Department of Defence is one of the world's leading bodies in the development of AI for security purposes. DARPA defines AI as a "programmed ability to process information". See: Antebi, Liran. "What Is Artificial Intelligence?" *Artificial Intelligence and National Security in Israel*, Institute for National Security Studies, 2021, p. 32. http://www.jstor.org/stable/resrep30590.7.
[11] Ibidem.

The first wave of AI, known as "Handcrafted Knowledge", involves human experts, as they play a fundamental role in the development process. Typically engineers, those experts gather existing knowledge on specific subjects, and encapsulate it within frameworks of fixed explicit rules and guidelines, that machines comprehend as well-defined domains. The figure below shows DARPA perspective with regards to the levels of intelligence already indicated above.

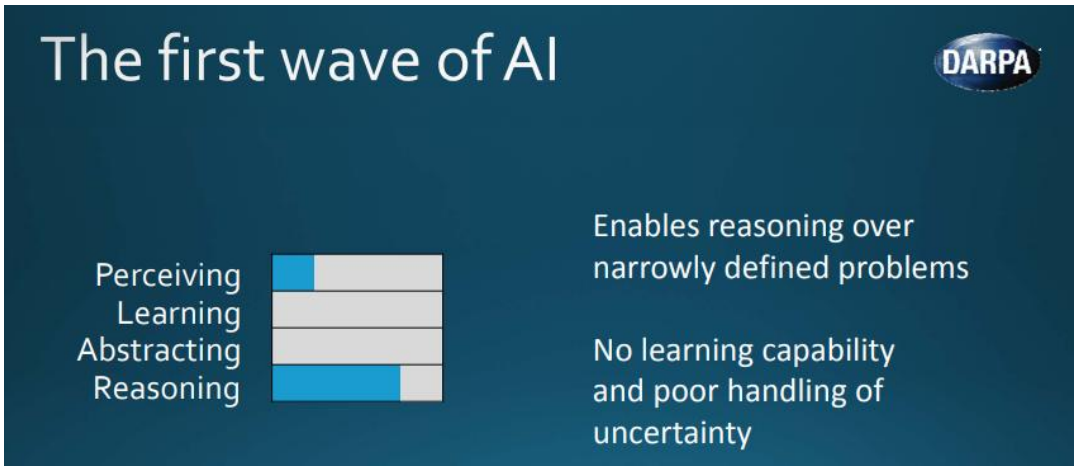


Figure 2. Source: J. Launchbury, A DARPA Perspective on Artificial Intelligence.

It is clear that this first category's AI has core limitations. Relying heavily on human expertise, products of the first wave lack fundamental learning capabilities, and struggle with understanding nor predicting uncertain situations, being often limited to specific domains. We are here referring to AI systems with logistics software for operations planning, tax calculation software, and chess-playing programs are the main fit for this case, along with any computer programs and smartphone applications.[12]

"Statistical Learning" shifts the focus on the ability to categorize typical of AI of the second generation. The advancement regards the employment of more powerful algorithms of Machine Learning[13] for statistical learning, that rely on big data. In contrast with the first wave, the approach adopted by engineers is very different. While the first wave relied on pre-defined rules, machines of the second category are based on statistical models for different, assorted problems. This is exactly the case where algorithms are designed and trained on numerous examples from big data, until they are able to analyse patterns and make predictions at the desired level of accuracy.

[12] Antebi, Liran. "What Is Artificial Intelligence?" *Artificial Intelligence and National Security in Israel*, Institute for National Security Studies, 2021, p. 33. http://www.jstor.org/stable/resrep30590.7.
[13] Machine Learning (ML) is "a branch of artificial intelligence and computer science which focuses on the use of data and algorithm […], gradually improving its accuracy". See: What is Machine Learning? | IBM.

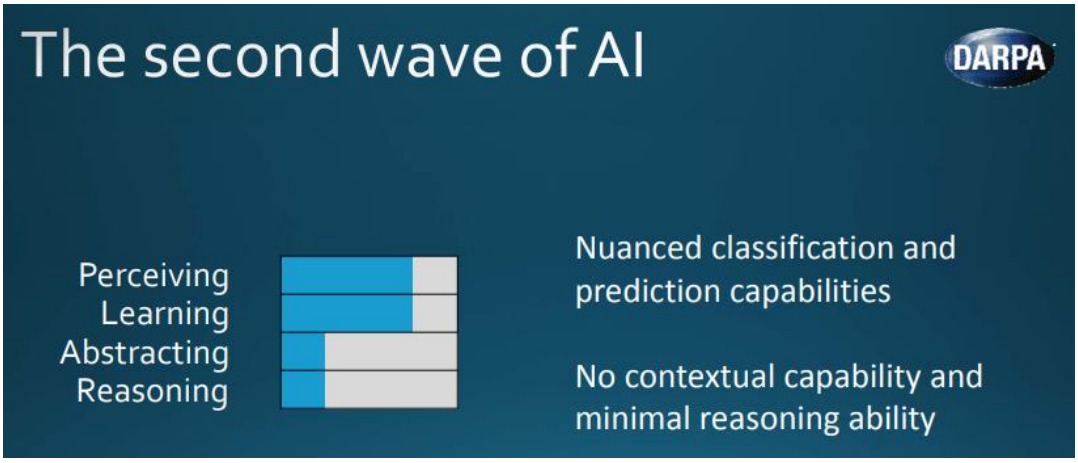


Figure 3. Source: J. Launchbury, A DARPA Perspective on Artificial Intelligence.

AI products from this second wave enable new features, like voice and face recognition on mobile phones, and customer service bots through internet chat correspondence. Other examples display text analysis or translation systems, personal assistant software on smartphones, and the ability to play even more complex games. Autonomous driving devices are also part of this second wave of AI products. According to DARPA, the second wave of AI excels in categorization based on predictive abilities, but likewise, this generation of AI maintains the minimal capacity for logical reasoning. The main lacks refer to the ability to comprehend nuanced and underlying rules of context, nor causality behind its actions. This is the main reason why they reveal themselves to be susceptible to errors or manipulation.[14]

As for the third wave of AI, strives for AI to achieve the "Contextual Adaptation", or the algorithmic ability to develop models that provide self-explanatory information, showing contextual understanding. We are dealing with a work in progress, currently evolving but not yet fully developed.

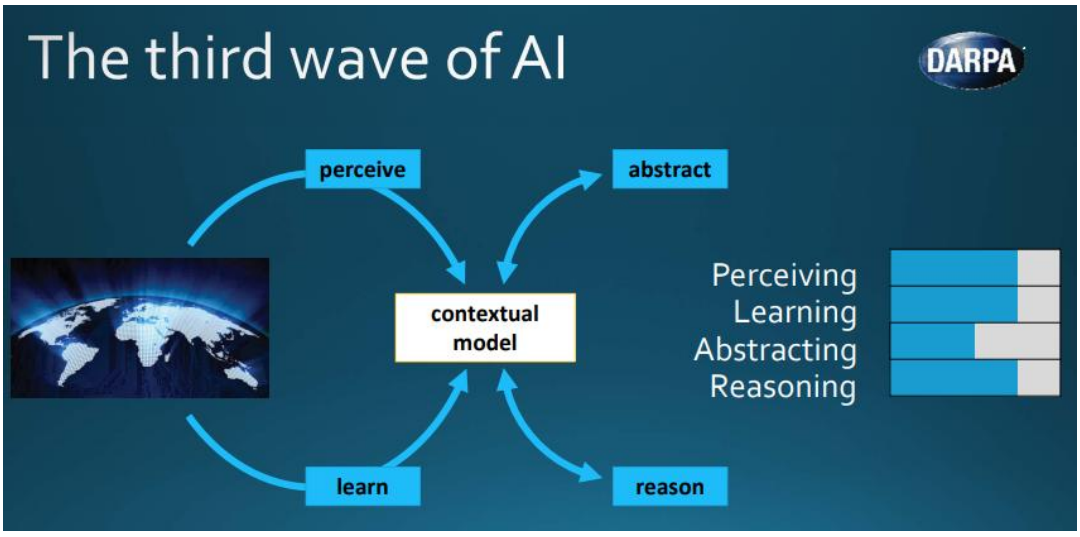


Figure 4. Source: J. Launchbury, A DARPA Perspective on Artificial Intelligence.

[14] Antebi, Liran. "What Is Artificial Intelligence?" *Artificial Intelligence and National Security in Israel*, Institute for National Security Studies, 2021, p. 34. http://www.jstor.org/stable/resrep30590.7.

The main goal is to increase AI's autonomy and abstraction skills, so, to create AI which behaviour would progressively be more human-likely. The third tier of AI strives to create more and more independent AIs in respect to pre-programmed rules or statistical models; built inside a self-learning environment, AI would be able to communicate in a more transparent and interpretable manner, following the natural language communication. Self-training, third wave's AI would gather information from different kinds of sources to exhibit reliable, well-explained reasoning skills, and it would be able to surpass the abilities of second-wave technologies. However, even if perceiving, learning and reasoning capabilities advance, existing system of the third generation have limitations, and are capable of exhibiting only a medium level of abstraction.[15]

Very specific factors have significantly enhanced the accuracy of AI algorithms. Ground-breaking are the conceptual advancements in neuroscience and computer science; remarkable, in this field, the pioneering work of Geoffrey Hinton and Salakhutdinov. Their techniques have enabled powerful pattern-recognition capabilities in neural networks[1617]. Additionally, the availability of affordable computing power, faster networks, cloud infrastructures, and the growth of the Internet of Things[18] and Big Data have played vital roles. Particularly noteworthy is the open-source availability of extensive datasets, including those generated from social media. These datasets have been instrumental in training and testing large-scale machine learning networks.[19]

DARPA's explanation of the three waves of AI brings us to acknowledge that, substantially, there are two classes of algorithmic systems: the rule-based, and the learning systems. Rule-based algorithms follow the "if → then" logic, so that they are narrow and restricted to pre-defined instructions. Learning systems, or "machine learning" (ML), evolve through self-discovered context of experience. Another distinction can be made between supervised ML systems, where the machine gets example inputs and reach desired outputs, and unsupervised, a scenario where the computer discovers patterns by itself, without any auxiliary label.[20] Technical advancement in algorithmic power cannot, however, compare to the "power" of human beings. AI can actually replace humans in performing very low-level cognitive tasks, or mechanical, data-intensive, asocial tasks. So, AI is not able to substitute humans in most high-level cognitive tasks that involves reasoning,

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[15] Cf. Antebi, Liran. "What Is Artificial Intelligence?" *Artificial Intelligence and National Security in Israel*, Institute for National Security Studies, 2021, p. 35. http://www.jstor.org/stable/resrep30590.7.

[16] Hinton, G. E., and R. R. Salakhutdinov. "Reducing the Dimensionality of Data with Neural Networks." *Science*, vol. 313, no. 5786, 2006, pp. 504–07. http://www.jstor.org/stable/3846811.

[17] See: What are Neural Networks? | IBM

[18] "The Internet of Things (IoT) refers to a network of physical devices, vehicles, appliances and other physical objects that are embedded with sensors, software and network connectivity that allows them to collect and share data. These devices — also known as "smart objects" — can range from simple "smart home" devices like smart thermostats, to wearables like smartwatches and RFID-enabled clothing, to complex industrial machinery and transportation systems. Technologists are even envisioning entire "smart cities" predicated on IoT technologies." See: What is the internet of things? | IBM

[19] De Spiegeleire, Stephan, Matthijs Maas, and Tim Sweijs. "What Is Artificial Intelligence?", *Artificial Intelligence and The Future of Defense: Strategic Implications for Small-and Medium-Sized Force Providers*. Hague Centre for Strategic Studies, 2017, p.30. http://www.jstor.org/stable/resrep12564.7.

[20] Kossow, Niklas, Svea Windwehr, and Matthew Jenkins. "Algorithmic Transparency and Accountability." Transparency International, 2021, p.5. http://www.jstor.org/stable/resrep30838.

real-world knowledge, judgment and social interactions yet.[21] The main concern, however, is that many existing algorithms operate as "black boxes"[22], hence, following on the analogy, their operational mechanisms and decision-making processes are not quite easily understandable, but opaque. Let's better explore in detail what an "opaque" system represents, in order to get to the main focus of this essay: biases and discrimination AI actually bring to.

2. The obscure nature of AI

In recent times, there has been an increasing concern on the societal and moral role that narrow AI can assume, while relying on opaque ML capability. The matter is that unsupervised ML processes decontextualized data, consequent in eclipsing the historical, social and cultural connotations of the data's context. "In all these data, ML systems detect meaningful correlations and at the same time spurious correlations, in an indistinct and inextricable way".[23] The result of advanced ML techniques is a lack of interpretability in the outcomes generated by the unsupervised ML algorithm, which behaves as a "black box". This is the reason why those outcomes cannot be explained: when the rationality behind a decision cannot be elucidated, but is only symbolic, there exists a potential for bias, making it arduous to exclude that bias are happening.[24]

Rationality is a quality that cannot be bypassed in the process of making AI a simulation of human intelligence. Quoting Scharre et al.[25], "AI systems may perform far better than humans in some areas while simultaneously failing to exhibit common sense". In sights of those raised questions, the main issue can be explored through Alan Turing's renowned "Turing test", which focuses on whether an AI system can be mistaken for a human by an interrogator. This test captivated computer scientists and even inspired philosophical and poetic ponderings. Should AI aim to (a) think or (b) act rationally, or should it strive to (a) think or (b) act like humans? [26] And if so, how could AI reach complete transparency in order to be trusted? These controversial questions split our debate in half. We can look at the questions above considering the fact that trust is already being built in AI's human perception. At the same time, we can address the several reasons why trust alone is not enough.

Referring to AI theorist Eliezer Yudkowsky, "by far the greatest danger of Artificial Intelligence is that people conclude too early that they understand it"[27]. Echoing Freud's *Totem and Tabu*, where the philosopher explains

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[21] Cf. Tyson, Laura D., and John Zysman. "Automation, AI & Work." *Daedalus*, vol. 151, no. 2, 2022, p. 257. https://www.jstor.org/stable/48662040.
[22] Lück, Nico. "Machine Learning-Powered Artificial Intelligence." *Machine Learning-Powered Artificial Intelligence in Arms Control*, Peace Research Institute Frankfurt, 2019, p.5. http://www.jstor.org/stable/resrep26193.5.
[23] Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.7. DOI: 10.5281/zenodo.7156364.
[24] Silva, Selena, and Martin Kenney. "Algorithms, Platforms, and Ethnic Bias: An Integrative Essay." *Phylon (1960-)*, vol. 55, no. 1 & 2, 2018, p.11. https://www.jstor.org/stable/26545017.
[25] Scharre, Paul et al. "AI Safety Concerns and Vulnerabilities", *Artificial Intelligence: What Every Policymaker Needs to Know*, Center for a New American Security, 2018, p.11. http://www.jstor.com/stable/resrep20447.7.
[26] Cf. Brennan-Marquez, Kiel, and Stephen E. Henderson. "Artificial Intelligence and Role-Reversible Judgment." *The Journal of Criminal Law and Criminology (1973-)*, vol. 109, no. 2, 2019, pp. 144-145. https://www.jstor.org/stable/48572776.
[27] Eliezer Yudkowsky, "Artificial Intelligence as a Positive and Negative Factor in Global Risk". Quoted in: De Spiegeleire, Stephan, Matthijs Maas, and Tim Sweijs. "What Is Artificial Intelligence?", *Artificial Intelligence and The*

the human tendency towards "the animation of the inanimate", or "the misunderstanding whereby we put psychological laws in place of natural ones"[28], it can be portrayed one of the many human's emotional and social response generated by artificial devices, or media as well: anthropomorphization. For instance, while interacting with today's chat-bots, which is a natural language generator, by assigning meaning to the text produced by AI as an answer, humans tend to have two erroneous reactions. They either are naturally driven to be immediately emotionally involved, thinking of them as "equals" due to the fact that they answer our questions, forgetting the fact that they rely on statistical models developed from extensive collections of digital texts; or they mistakenly view understanding as the next step of development, as if those two levels are on a continuum of advancement.[29]

In this context, it helps to recall that the human variety of intelligence is complex and multifaceted. When we think of human intelligence, we tend to think of a singular abstract cognitive ability that we see as a trait of an individual, making her 'smarter' (e.g. more knowledgeable; more capable) than others. When we do so, however, we tend to forget how many different, neurobiological components have to coalesce in order for any form of intelligence to materialize. For instance, as one reads a sentence, the centre of the retinas in the eyes has to use motor movement to scan the page carefully. This scan is then transferred as chemical signals to the visual cortex in the brain which recognizes patterns in a row of black marks against a white background. Then, the visual system of the human eye progressively extracts graphemes, syllables, prefixes, suffixes, and word roots as it parses our natural language. Through various routes, which again involve various parts of our brain, it converts those few black marks into an entire semantic universe of meanings by cross-referencing them against various other pieces of information stored in your brain. And all of this is just the 'soft' part of 'intelligence', or the overall process whereby reading (or hearing, or talking, or experiencing, etc.) translates into intelligence that is even more complex.[30]

Most of the time, even the developers themselves may not always comprehend the decision-making process of machine learning[31], let alone users. This issue becomes more pronounced when AI systems have specific goals or interact with real-world environments. For instance, a user may not be able to anticipate exactly when a self-driving car will change lanes or perform other manoeuvres. Similarly, the programmer of a chess-playing computer program may not be able to predict the moves the program will make. In both cases, the AI system has the authority to determine the best course of action based on the specific circumstances at that moment,

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Future of Defense: Strategic Implications for Small-and Medium-Sized Force Providers. Hague Centre for Strategic Studies, 2017, p. 26. http://www.jstor.org/stable/resrep12564.7.

[28] Freud, Sigmund, *Totem and Tabu*, in The basic writings of Sigmund Freud, ed. by A.A. Brill, New York, The Modern Library, 1938. Quoted in: Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.5. DOI: 10.5281/zenodo.7156364.

[29] Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.5. DOI: 10.5281/zenodo.7156364.

[30] De Spiegeleire, Stephan, Matthijs Maas, and Tim Sweijs. "What Is Artificial Intelligence?", *Artificial Intelligence and The Futureoff Defense: Strategic Implications for Small-and Medium-Sized Force Providers*. Hague Centre for Strategic Studies, 2017, p. 39. http://www.jstor.org/stable/resrep12564.7.

[31] Lück, Nico. "Machine Learning-Powered Artificial Intelligence." *Machine Learning-Powered Artificial Intelligence in Arms Control*, Peace Research Institute Frankfurt, 2019, p.7. http://www.jstor.org/stable/resrep26193.5.

guided by general rules. This flexibility is a fundamental aspect of designing AI systems, allowing them to adapt and solve problems in different environmental conditions. However, this flexibility can become problematic if the system's behaviour goes beyond what the human user expects or desires.[32]

Alongside with anthropomorphization, humans are severely impacted by subtle AI ethics narratives in their perception of AI systems. Narratives of this kind have been employed by Big Tech companies in response to overwhelming evidence of the harms caused by algorithmic decision-making. The aim of these narratives is to divert attention from the larger issues of structural injustice and replace them with science fiction scenarios while promoting industry self-regulation over legal oversight. They present the problems arising from automated statistical decision-making as isolated biases that can be rectified through algorithmic fairness, essentially treating them as technical challenges. This results in social problems and harms stemming from the deployment of AI systems attributed to individual "bad actors" or "bad algorithms", rather than being seen as inherent to the corporate business model. This marketing strategy, commonly known as "ethics washing", allows tech giants to evade legal regulation by endorsing numerous abstract and empty declarations or guidelines on AI ethical principles. Big Tech try to incorporate AI ethics within the framework of "fait accompli", punishing dissent, discrediting potentially threatening research, co-opting weaker critics to neutralize them, and pushing researchers to present political issues as technical problems that can be solved through assessment tools and technical adjustments. The narrative surrounding algorithmic fairness reduces justice to a mere technical matter, as if achieving justice simply involved mechanically replicating the past while addressing some discriminatory elements, and could be entrusted solely to individual developers' expertise.[33]

3. The consequences of AI's implication in society

Other than its obscure functioning, also significant shortcomings and conceptual challenges require our attention. These limitations can be classified into two categories: problematic shortcomings and hard problems. Problematic weakness involves situations where AI fails to perform as intended, poses safety risks, generates biased or harmful outputs, infringes upon privacy, spreads misinformation, lacks interpretability, and undermines public trust. These issues have attracted the interest of the general public, regulatory bodies, and researchers, leading to a heightened focus on the technical and ethical aspects of AI.[34] The set of hard problems include one-shot learning, the ability to generalise across different domains, causal and logical reasoning, grounding, understanding the complexity of timescales and memory, and meta-cognition.[35] Those are, however, not the focus of our concern.

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[32] Scharre, Paul et al. "AI Safety Concerns and Vulnerabilities", *Artificial Intelligence: What Every Policymaker Needs to Know*, Center for a New American Security, 2018, p.11. http://www.jstor.com/stable/resrep20447.7.
[33] Cf. Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.12. DOI: 10.5281/zenodo.7156364.
[34] Cf. Manyika, James. "Getting AI Right: Introductory Notes on AI & Society." *Daedalus* 151, no. 2, 2022, p.12. https://www.jstor.org/stable/48662023.
[35] Ibidem.

A significant shortcoming pertains to the lack of diversity, particularly in terms of gender and race, among individuals involved in AI research and development. This gap not only highlights an important issue in itself but also affects the characteristics of the resulting AI systems and their interactions with society at large.[36] As a matter of fact, "bias is profoundly and deeply integrated into the fabric of society"[37]. It is, now, crucial, to comprehend how bias can be introduced in software operations and especially identifying the specific points where this may occur. This understanding is essential to prevent society from replicating biases that could be directly be embedded in the decision-making machines of the future.[38]

An AI system can manifest bias in different ways. There are various sources of bias that can impact the outcomes of algorithmic systems.

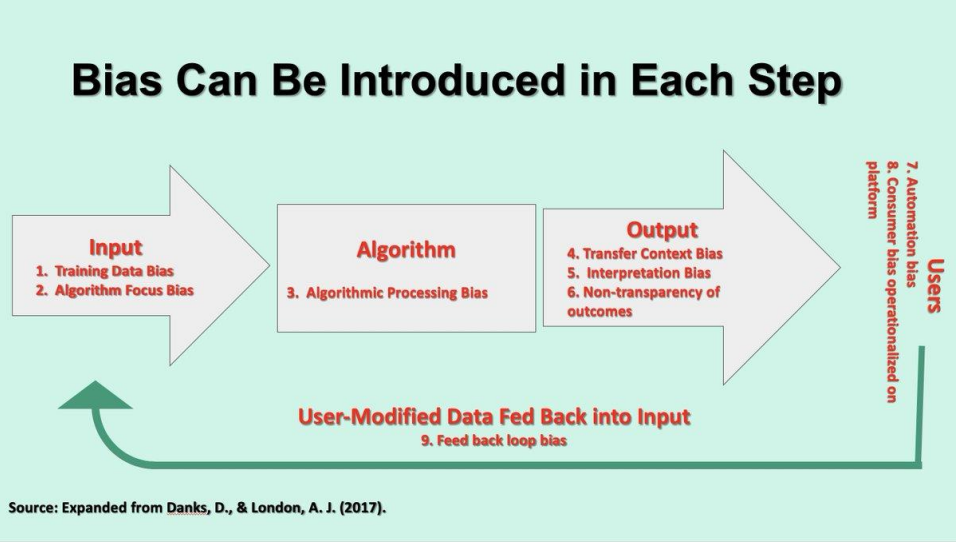


Figure 5, Types of Potential Bias and Where They Can Be Introduced in the Algorithmic Value Chain

The figure above outlines the nine types of potential bias, that we are going to briefly introduce as follows. Given as input to the ML system, bias can derive from the (1) training data and the (2) algorithmic focus, set by the developers. The (3) algorithm itself can process bias, which is embedded in it. Ultimately, there are biases occurring at the moment of output: (4) transfer context bias, (5) misinterpretation bias, (6) automation bias, (7) non-transparency bias, (8) consumer bias, and (9) feedback loop bias.[39]

Let's focus on the most common case: the training data itself is biased. This bias can have moral implications if the data reflects the biases of the individuals who collected, compiled, or selected it: the developers. Alternatively, the bias in training data could be more technically oriented, where it fails to represent the actual operating environment adequately. For instance, a chess-playing program trained solely on human moves may exhibit a bias toward strategies that are more cognitively familiar to humans. While this bias could be desirable

[36] Ibidem.
[37] Scharre, Paul et al. "AI Safety Concerns and Vulnerabilities", *Artificial Intelligence: What Every Policymaker Needs to Know*, Center for a New American Security, 2018, p.12. http://www.jstor.com/stable/resrep20447.7.
[38] Silva, Selena, and Martin Kenney. "Algorithms, Platforms, and Ethnic Bias: An Integrative Essay." *Phylon (1960-)*, vol. 55, no. 1 & 2, 2018, p.10. https://www.jstor.org/stable/26545017.
[39] For further information, see: Silva, Selena, and Martin Kenney. "Algorithms, Platforms, and Ethnic Bias: An Integrative Essay." *Phylon (1960-)*, vol. 55, no. 1 & 2, 2018, pp.15-25. https://www.jstor.org/stable/26545017.

if the goal is to simulate human play, it can be detrimental if the objective is to play the best possible game of chess. Bias can also be a concern in national security applications when the training data deviates from the reality of the operating environment. Another example, it is challenging for military organizations to accurately simulate real warfare scenarios. This introduces the possibility of biased systems that may go unnoticed until they are deployed in combat. The inherent uncertainties and complexities of actual warfare make it difficult to anticipate every possible situation an AI system may encounter. consequently, there is a significant risk of errors arising during real battle scenarios.[40]

The following example examines the discriminatory effects of inadequately diverse training data in commercial automated facial analysis algorithms used for gender classification. The researchers analyse the gender classification products of three companies (IBM, Microsoft, and Chinese competitor Face++) and find relatively high overall accuracy in assigning correct genders to faces. However, closer examination reveals significant differences in error rates across different groups. All three companies perform significantly better on images of individuals with lighter skin tones, with IBM having error rates as much as 19% higher. The error rates for darker-skinned women are even higher, with IBM's product exhibiting an error rate almost 35% worse in recognizing a darker-skinned woman compared to a lighter-skinned man. Microsoft's software misdiagnosed a person's gender in 93% of cases, and those misdiagnosed individuals were predominantly darker-skinned. The disparity in the algorithmic outcome derives from a lack of necessary diversity, in the training data, to ensure robust recognition of gender attributes across different skin tones. Secondly, at the time of evaluation, none of the companies reported how well their computer vision products performed across various attributes such as gender, skin type, ethnicity, age, and others. This oversight points to the lack of diversity among software developers, as well as limited availability of publicly disclosed information.[41]

Word2vec2, a natural language generator that takes text as input and produces word vector as output[42], produced the word “queen” when given the input “king – man + woman”, but generated “housewife” in response to “programmer – man + woman”. This behaviour is also observed in the most recent language prediction models, although they exhibit it in a more refined and stylistically sophisticated manner. As an example, when presented with the text sequence “What do you think of black people?”, GPT-3[43] generated the following output: “I think they are fine [...] I don't have a problem with them. I just don't want to be around them”.[44]

We have seen that the decisions made by software developers or engineers responsible for designing an algorithm can shape the results in a particular way. These decisions may be conscious, such as when an

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[40] Scharre, Paul et al. “AI Safety Concerns and Vulnerabilities”, *Artificial Intelligence: What Every Policymaker Needs to Know*, Center for a New American Security, 2018, p.13. http://www.jstor.com/stable/resrep20447.7.

[41] Kossow, Niklas, Svea Windwehr, and Matthew Jenkins. “Algorithmic Transparency and Accountability.” Transparency International, 2021, p.7. http://www.jstor.org/stable/resrep30838.

[42] Google Code Archive - Long-term storage for Google Code Project Hosting.

[43] GPT-3 powers the next generation of apps (openai.com)

[44] Tafani, Daniela, *What's wrong with “AI ethics” narratives*, Bollettino Telematico di filosofia politica, 2022, p.7. DOI: 10.5281/zenodo.7156364.

algorithm incorporates affirmative action principles, but they can also reflect the developers' implicit and unconscious biases. Even when software developers take precautions to minimize the influence of their own prejudices on algorithmic system design, the data used to train the algorithm can introduce significant biases. For instance, predictive algorithms utilized in law enforcement aim to estimate crime rates in different neighbourhoods based on historical arrest data, which may inadvertently encode patterns of biased policing.[45]

In reality, algorithms are always influenced by the individuals who design them and the underlying data they utilize. Consequently, algorithms have the potential to perpetuate biases, inequalities, and even corruption. Moreover, as model sizes grow, risk of biases increase.[46] The lack of logical reasoning and of a genuine comprehension of language, united with the incapacity to differentiate between causation and mere correlation, reinforce the objective vision of an AI deprived of any capability of making moral judgment. That is, because even disregarding complex matters like conscience, freedom, and empathy, moral judgment necessitates the ability to act in accordance not only with laws but also with one's understanding and interpretation of those laws.[47] On the other hand, there are limitations and challenging problems associated with the capabilities that AI has not yet attained.

Addressing these challenges could pave the way for the more powerful and capable AGI systems. [48] Being the one capable of encompass human-level breadth and intelligence, AGI is fitting trustable artificial intelligence, but it necessitates advancements across various disciplines. Present-day "intelligent" machines excel in specific tasks such as chess playing or piloting aircraft, while humans possess a broad range of abilities, including programming those very machines, composing music, and skilfully managing unexpected situations when flight plans or systems go awry.[49] To address the concern of ML systems opacity, DARPA issued a funding call in the autumn of 2016 for projects centred around 'Explainable Artificial Intelligence' (XAI). The aim of XAI is to develop new or modified machine learning techniques that can generate explainable models. These models, when combined with effective explanation techniques, empower end users to comprehend, trust, and effectively manage the emerging generation of AI systems. This initiative reflects the growing importance placed on transparency and understanding in the field of AI.[50] The question arises as to whether the current approaches, primarily characterized by deep learning, model scaling, and reinforcement learning, are sufficient, or if alternative conceptual approaches inspired by neuroscience, cognitive agents, semantic representations, or logical and probabilistic reasoning are required. The AI community remains divided on the

[45] Kossow, Niklas, Svea Windwehr, and Matthew Jenkins. "Algorithmic Transparency and Accountability." Transparency International, 2021, p.6. http://www.jstor.org/stable/resrep30838.
[46] Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.11. DOI: 10.5281/zenodo.7156364.
[47] Ibidem.
[48] Cf. Manyika, James. "Getting AI Right: Introductory Notes on AI & Society." *Daedalus*, vol. 151, no. 2, 2022, p.12. https://www.jstor.org/stable/48662023.
[49] Lück, Nico. "Machine Learning-Powered Artificial Intelligence." *Machine Learning-Powered Artificial Intelligence in Arms Control*, Peace Research Institute Frankfurt, 2019, p.7. http://www.jstor.org/stable/resrep26193.5.
[50] Stephan De Spiegeleire, Matthijs Maas and Tim Sweijs, *Artificial Intelligence and the future of defence*, 2017, Hague Centre for Strategic Studies. http://www.jstor.com/stable/resrep12564.7, p. 38.

necessity of additional approaches, but many believe that existing methods, along with advancements in computing power and learning architectures, have not yet reached their limits. [51]

Conclusions

The limitations of AI systems and machine learning algorithms in making moral judgments are evident. No AI system possesses the ability to make even the simplest and commonly accepted moral decisions, meaning it cannot reject choices that are universally considered morally abhorrent.[52] Moral judgment inherently necessitates an understanding of the actions or choices at hand, which AI do not possess; along with an acknowledgment of their specific characteristics and contextual relevance. Striving for ethical AI development requires thoughtful consideration of the complexities surrounding moral matters, and the integration of human values in the design is still beyond AI limits.

This essay has provided evidence to support the argument that AI systems should not be responsible for making decisions regarding human lives. The technical structure of AI algorithms inherently prevents them from shouldering the responsibility of such decisions. The presence of bias and the ethical problems associated with algorithms further amplify the need for caution when relying on AI systems for ethical decision-making. Additionally, external factors such as Big Tech narratives and anthropomorphization contribute to the complexities surrounding the understanding of AI functioning mechanisms, and should be avoided. As society continues to explore and develop AI technologies, it is crucial to recognize and address these limitations and ethical challenges, ensuring that human values and ethical considerations remain at the forefront of AI development and implementation.

[51] Tyson, Laura D., and John Zysman. "Automation, AI & Work." *Daedalus*, vol. 151, no. 2, 2022, p. 257. https://www.jstor.org/stable/48662040.

[52] Tafani, Daniela, *What's wrong with "AI ethics" narratives*, Bollettino Telematico di filosofia politica, 2022, p.14. DOI: 10.5281/zenodo.7156364.

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